Sehaj Randhir Singh

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Founder, Guerilla · QHSS Class of 2028

I build learned control for physical machines that comes with proofs. My startup, Guerilla, is building the Index of Operations: one representation that reads a machine as the shape its physical parts draw together, and uses that shape to monitor it, model its dynamics and control it.

FOUNDER

Guerilla — the Index of Operations (IOO)
2026 – Present · Founder · solo
An index of the physical operators machines are made of (motors, pumps, joints, thermal masses), composed where they pass energy and force, so a machine's telemetry becomes a shape. Invariant to sensor recalibration by construction: on ESA's OPS-SAT satellite telemetry it holds at 0.749 → 0.736 AUROC while raw features fall to 0.487, and it lifts the published OPS-SAT baseline from 0.911 to 0.951. The same signal that detects a fault gates the controller, cutting tracking error 8.2× under a 5× actuator fault in simulation. Tested on satellite, jet-engine and hexapod-robot data.

RESEARCH & INDUSTRY

NYU Tandon ECE — Wenqi Cui Research Group
2026 – Present · Research Collaborator · first-author paper accepted, NeurIPS 2026 workshop
Certified neural Lyapunov functions for power-grid transient stability. Showed a learned certificate passing 99.98% of samples fails a formal verifier (0 of 200,000 random samples vs 6.1% under directed search), then certified training through the CROWN bound to ρ = 2.5 on all five seeds vs 1.8 for counterexample repair. Only group member working on Lyapunov stability.

Research Intern — Data-Center Grid Infrastructure
2025 – Present · confidential
Researching power-distribution bottlenecks and grid strain driven by the compute demand of large-scale data centers, including generation-and-tie planning for how new load interconnects to the grid.

Power-Systems Modeling — Grid Infrastructure
Ongoing · confidential
Build and run power-flow simulations for grid infrastructure design and produce engineering reports that inform line selection and routing.

NYU Tandon MAE — Mechatronics, Controls, and Robotics Lab
2026 · with Prof. Vikram Kapila · Research Collaborator · manuscript in preparation
Motion-first multi-object re-identification for robotics: identity as a decision problem, with gait, trajectory and geometry as primary signals.

Antioch — Robotics Simulation
2025 – 26 · research collaboration
Control and stability for simulated physical systems with an early-stage robotics-simulation startup, contributing technical feedback and development.

PROJECT

The AI Data-Center Boom: Control and Systems Flexibility for a Grid Under AI Load, with OpenAI, Tesla Megapack, Mitsubishi Heavy Industries, NYISO, PJM, ERCOT and NYSERDA
2026 – Present · independent study
How the grid can take on AI data-center load through control instead of only new build-out: data centers and batteries as flexible, controllable loads, faster certified interconnection screening, and frequency and voltage margins under fast-ramping demand.

PUBLICATIONS & MANUSCRIPTS

Learning Lyapunov Functions for Power System Transient Stability: Neural Verification and Retraining. S. R. Singh, W. Cui. *Tackling Climate Change with ML Workshop, NeurIPS 2026.* **Accepted, camera-ready in preparation.**

Two manuscripts submitted to IEEE. 2026. **Under review.**

Geometric State Manifolds for Transferable, Data-Efficient Dynamic-System Learning. Manuscript, 2026.

The Shapes a Machine Draws: Recalibration-Invariant Relational Atoms of Multi-Physics Operation. Manuscript, 2026, 22 pp.

Time Is Geometry: A Scale-Covariant Operator Model for Dynamical Systems. Manuscript, 2026.

Certified Feasible-Capacity Operators for Interconnection Queues. Manuscript, 2026.

Certify, Then Switch: A Physics-Informed Operator for Grid Reconfiguration Under Out-of-Distribution Data-Center Loads. Manuscript, 2026.

Certifying Causal Scaling: A Metamorphic Verifier for Learned Physical Predictors. Manuscript, 2026.

Rank-Fidelity-Gated Optimization: Keeping Certificate Descent Safe for Learned System Components. Manuscript, 2026.

Plus 8 more manuscripts on certified grid control and learned operators, listed at sehajr-singhs.github.io.

SELECTED BUILDS

Operating Atlas — recalibration-invariant machine fingerprints
2026 · solo · github.com/sehajr-singhs/operating-atlas
Nine descriptors of the shapes a machine's channels trace identify 7 real systems at 99.4%, and keep 99.0% under re-instrumentation where marginal statistics fall to 80.3%.

Time-Series Operator — time as geometry, not tokens
2026 · solo · github.com/sehajr-singhs/time-series-operator
Delay embedding, Koopman lifting and a learned vector field pretrained across domains. 31 of 40 zero-shot wins against a per-series GRU, and recovers the 11-year solar cycle from a held-out channel.

Stability-Constrained Neural Adaptive Control
2025 – 26 · solo · github.com/sehajr-singhs/adaptive-contact-dynamics
Bounded RBF network inside a Lyapunov-derived adaptive law. 2.06° vs 98.98° joint error against the unconstrained ablation on a UR5e contact task in MuJoCo.

Operator — switched-certificate control under wear
2026 · solo · MIT research competition semifinalist
A learned controller for a degrading electromechanical plant that carries a per-step control-Lyapunov certificate through the whole degradation.

Steerable VLA — hierarchical vision-language-action with a safety envelope
2026 · solo · github.com/sehajr-singhs/steerable-vla
VLM subgoal planner, flow-matching action expert, a steerable conditioning layer with a certified no-jerk bound and runtime verification, implemented at miniature scale on cable untangling.

COMPETITIONS & RECOGNITION

MIT Research Competition — Semifinalist *2026 · top 29 of 1,042 applicants (~3%)*

Anduril AI Grand Prix — Team GREMLIN, Team Lead & Algorithm R&D Lead *2026 · Qualifier 1 passed · autonomous drone racing*

OpenAI Parameter Golf Challenge — Advanced Development Grant *2026*

Reviewer — ML4PS Workshop, NeurIPS 2026 *2026*

MIT Beaver Works CubeSat Program — Co-Lead & Engineering Lead *Sept 2025 – Present*

FTC Robotics — Team 19426, World Championship *2025 – 26 · mechanical, CAD & strategy*

QHSS HACK! — Hack Hardware Lead *2026 – Present · founded the school's engineering club, 2025*

EDUCATION

Queens High School for the Sciences at York College (QHSS) *Class of 2028 · Advanced Research track*

Columbia University — Energy Systems Coursework (Powering the Future) *2026 – 27*

SKILLS

Control & dynamics	Lyapunov & ISS stability, MRAC, MPC, contraction, region-of-attraction certification, Koopman/DMD
ML & verification	PyTorch, JAX, graph neural networks, neural ODEs, auto_LiRPA / α,β-CROWN, dReal, MuJoCo
Systems	Python, C/C++, MATLAB, power-flow simulation, Git, LaTeX, CAD, CNC, 3D printing